

Face Detection, Tracking, and Classification from Large-Scale News Archives for Analysis of Key Political Figures

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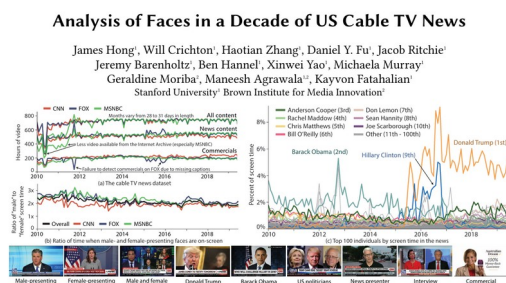
Why this matters ?

- TV news content analysis still relies heavily on human coding.
- Today's media is increasingly video-centric; we need scalable, objective measures like on-screen presence and screen time.
- Computer vision can reach human-level accuracy for “who appears when” at scale.

Research Question



- Can we build a high-accuracy, scalable method to detect, track, and classify key political faces in a long-running TV news archive and use it to study substantive media questions?



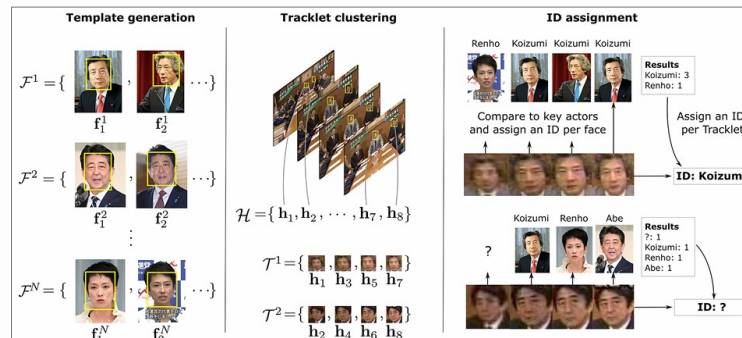
- Can we build a high-accuracy, scalable method to detect, track, and classify key political faces in long-running TV news archives **without retraining when adding new targets** and use it to study substantive media questions?



Contributions



- A new two-stage pipeline:
 1. Detect faces & embed them
 2. Cluster into **tracklets** and vote against template embeddings (no need for retraining).
- Works from 3–5 images per target.
- Validated on popular news channels. (3 U.S. cable channels and 2 Japanese newscasts)
- Code released to the community.



Data



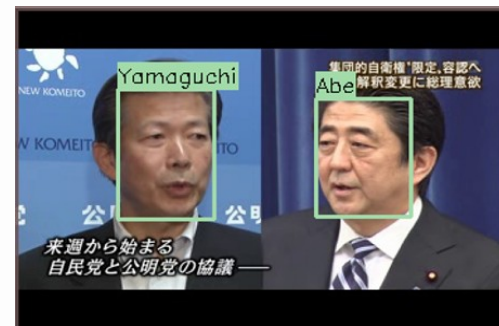
U.S. TV Archive:

- Source: CNN, FOX, MSNBC; frames sampled from 874 programs (~1,140 hours) with labels for 53 figures.
- Evaluation subset: 2012–2019 (due to access).
- U.S. frames: 4,346.



Japan TV Archive:

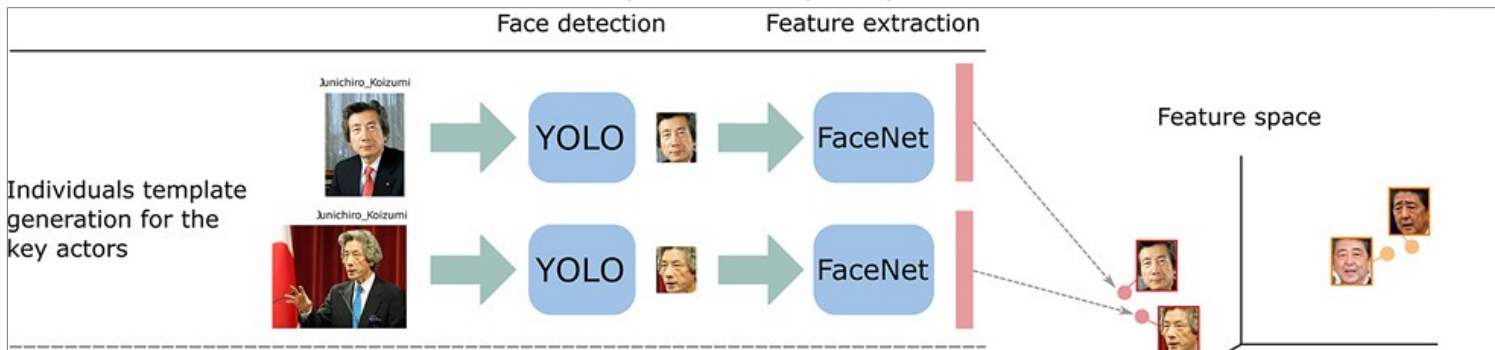
- NHK “News 7” and ANN “HODO Station”.
- 41 politicians (party leaders 2001–2021).
- 19,823 annotated frames, ~29k bounding boxes.



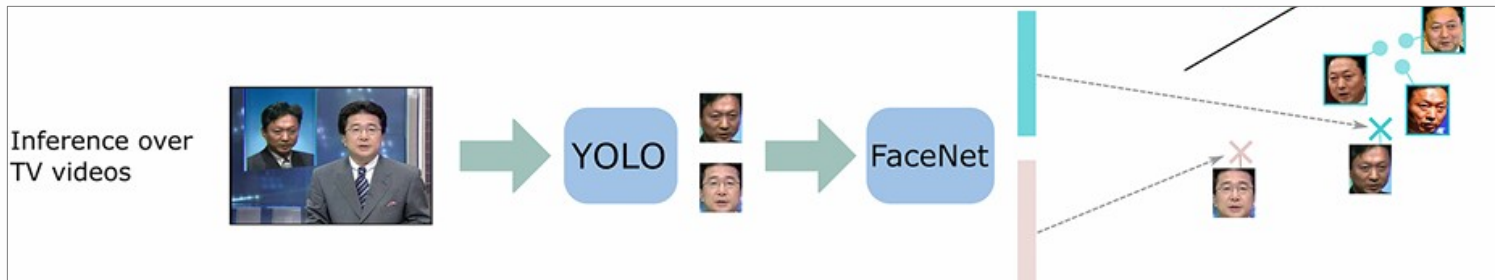
Pipeline



1. **Templates:** download 3–5 face images per target figure, extract 512-D FaceNet



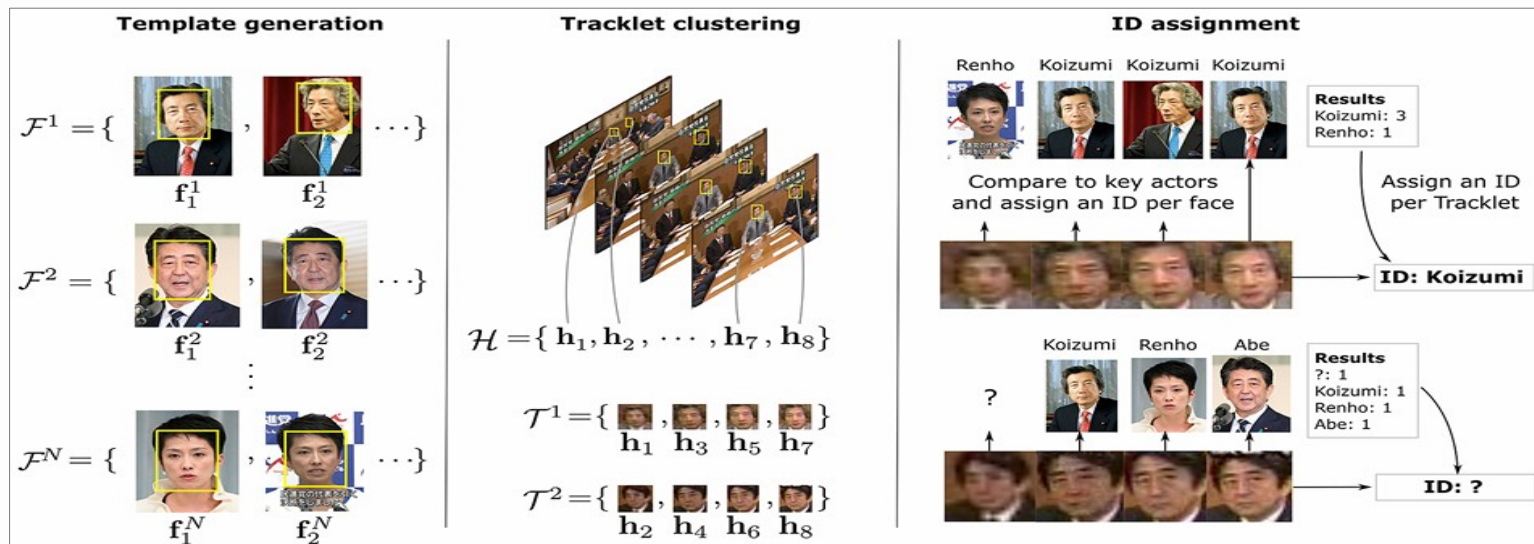
2. Inference over TV videos



Pipeline



3. **Tracklets:** cluster embeddings across adjacent frames to form tracklets.
4. **Classification:** for each tracklet, we perform a majority voting.



Results



Choice of detector:

Detector	CNN	FOX	MSNBC	NHK	HODO	OVERALL
DFSD	4.62%	3.88%	5.16%	3.13%	6.28%	4.61%
MTCNN	5.36%	4.60%	6.36%	2.84%	8.55%	5.54%
YOLO	1.11%	0.96%	1.01%	0.65%	0.33%	0.81%

Table 3. Missed detections percentage per TV channel over the three different face detectors DFSD, MTCNN, and YOLO-face.

Metrics:

$$P = \frac{TP}{TP + FP}, R = \frac{TP}{TP + FN}, F1 = 2 \cdot \frac{P \cdot R}{P + R}$$

Results (Japan)



Detector	Classifier	Overall			NHK News 7 (2013–2021)			HODO station (2014–2021)		
		P	R	F1	P	R	F1	P	R	F1
DFSD	KNN	0.81	0.69	0.738	0.81	0.75	0.776	0.80	0.63	0.701
MTCNN		0.78	0.69	0.731	0.79	0.74	0.766	0.77	0.63	0.696
YOLO		0.80	0.72	0.756	0.81	0.78	0.793	0.79	0.66	0.720
DFSD	Centroid	0.78	0.74	0.760	0.77	0.79	0.780	0.78	0.70	0.741
MTCNN		0.79	0.77	0.778	0.79	0.82	0.806	0.78	0.72	0.749
YOLO		0.77	0.79	0.785	0.79	0.84	0.816	0.75	0.75	0.754
DFSD	Vote	0.83	0.74	0.784	0.82	0.78	0.799	0.85	0.70	0.768
MTCNN		0.86	0.73	0.793	0.86	0.78	0.819	0.87	0.69	0.768
YOLO		0.85	0.78	0.815	0.84	0.81	0.825	0.87	0.75	0.804

Results



Tracking effect:

Detector	Tracking	CNN			FOX			MSNBC			NHK			HODO		
		P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
DFSD	No	1.0	0.65	0.787	1.0	0.67	0.804	1.0	0.61	0.761	0.79	0.75	0.769	0.78	0.63	0.700
DFSD	Yes	1.0	0.78	0.875	1.0	0.77	0.867	1.0	0.73	0.844	0.82	0.78	0.799	0.85	0.70	0.768
MTCNN	No	1.0	0.65	0.787	1.0	0.66	0.793	1.0	0.59	0.744	0.78	0.75	0.765	0.76	0.64	0.695
MTCNN	Yes	1.0	0.77	0.869	1.0	0.75	0.858	1.0	0.69	0.814	0.86	0.78	0.819	0.87	0.69	0.768
YOLO	No	1.0	0.70	0.827	1.0	0.71	0.830	1.0	0.66	0.797	0.80	0.78	0.791	0.78	0.67	0.722
YOLO	Yes	1.0	0.82	0.902	1.0	0.80	0.889	1.0	0.76	0.865	0.84	0.81	0.825	0.87	0.75	0.804

Table 6. Performance with or without face tracking for different detectors.

Applications

Table 7. Screen time of major U.S. candidates over the 847 randomly sampled videos for the 2016 general elections.

		CNN	FOX	MSNBC	Total
Republican primary (2016-02-01 to 2016-06-07)	Donald Trump	6,470	2,670	4,548	13,688
	Ted Cruz	3,454	2,330	3,522	9,306
	Marco Rubio	1,838	1,025	1,435	4,298
	John Kasich	1,206	1,246	1,596	4,048
Democratic primary (2015-04-12 to 2016-06-02)	Hillary Clinton	9,953	3,744	9,916	23,613
	Bernie Sanders	8,941	1,828	7,362	18,131
General election (2016-06-02 to 2016-11-08)	Donald Trump	5,064	2,204	6,425	13,693
	Hillary Clinton	3,562	2,476	6,497	12,535

Discussion



Upsides:

- Continuous, person-level visibility metrics for TV/video at national scale (e.g., per-party/per-candidate “share of screen”).
- Near-real-time dashboards during elections (daily/weekly fairness monitors).
- Newsrooms can self-monitor the balance and diversity of coverage.
- No retraining required.

Concerns:

- Campaigns could manipulate visibility (e.g., placing portraits or cutouts to spoof detectors).
- Non-targets (bystanders, staffers) may be captured.
- Code is released publicly without any license. Should be licensed as “License code” with research-only or no-surveillance clauses.

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- How do we weight TV vs. YouTube/TikTok/podcasts to estimate true exposure today?